

Hospital Emergency Department Operations & Patient Waiting Time Analysis

Introduction:

Emergency Departments (EDs) play a critical role in delivering timely healthcare services. However, increasing patient volumes, resource constraints, and operational inefficiencies often lead to prolonged waiting times and overcrowding. These challenges can negatively impact patient satisfaction, treatment outcomes, and overall hospital performance.

This project focuses on analyzing Emergency Department operations to identify key factors influencing patient waiting times and patient flow. Using a combination of Excel, SQL, Power BI, the project examines patient arrival patterns, department referrals, admission trends, and operational bottlenecks that contribute to delays in care delivery.

The analysis aims to provide actionable insights that can help hospital administrators optimize resource allocation, improve staffing decisions, and enhance operational efficiency. In addition to descriptive and diagnostic analytics, a 30-day waiting time forecast is performed using historical patient data to identify future trends and support proactive operational planning. This enables hospital management to anticipate potential demand fluctuations, allocate resources effectively, and improve patient experience.

Through interactive dashboards and data-driven analysis, this project demonstrates how healthcare organizations can leverage data to improve Emergency Department performance, reduce operational bottlenecks, and support informed decision-making for better healthcare delivery.

Business Problem Statement

Emergency Departments frequently experience high patient volumes, resulting in:

- Long waiting times before consultation
- Overcrowding during peak hours
- Uneven distribution of medical resources
- Delays in treatment and admission processes

Hospital management requires insights into:

1. Factors contributing to increased waiting times.
2. Peak patient arrival periods.
3. Departments causing operational delays.
4. Admission and discharge patterns.
5. Predicting patient waiting time before consultation.
6. Staffing optimization opportunities.

Project Objectives

1. Analyse patient arrival patterns and waiting times to evaluate Emergency Department performance.
2. Identify operational bottlenecks and factors contributing to patient delays.
3. Perform root cause analysis to understand trends affecting patient flow and resource utilization.
4. Forecast patient waiting times for the next 30 days using historical waiting time trends.
5. Generate actionable insights and recommendations to improve hospital efficiency, staffing, and patient experience.

Dataset Overview

Dataset used: Hospital Emergency Room Dataset

Source: Kaggle Dataset repository

The dataset used in this project contains **9,216 Emergency Department patient records**, capturing information related to patient visits, waiting times, and hospital operations. The data was used to analyse patient flow, identify factors contributing to delays, and evaluate Emergency Department performance.

The original dataset included the following attributes:

Column	Description
Patient ID	Unique identifier for each patient
Gender	Patient gender
Age	Patient age
Arrival Date	Date of patient arrival
Arrival Time	Time of patient arrival
Patient Wait Time	Time waited before receiving treatment
Patient Admin Flag	Indicates whether the patient was admitted (True/False)
Department Referral	Department assigned for consultation or treatment

The dataset provides a realistic representation of Emergency Department operations, allowing for the analysis of patient arrival patterns, waiting time trends, admission rates, and departmental workloads. Additional analytical columns such as Arrival Hour, Severity Category, and Waiting Risk Category were created during the data preparation stage to support deeper analysis and visualization.

Methodology → Result

1.Data Preparation and Cleaning

Objective: Prepare the raw dataset for analysis by improving data quality and creating additional analytical features.

- Checked the dataset for missing values and inconsistencies.
- Validated data formats and standardized column structures.
- Created derived fields including **Arrival Hour**, **Age Group**, **Severity Level**, and **Waiting Risk Category** to support deeper analysis.
- Prepared a clean and structured dataset for SQL analysis and Power BI visualization.

Outcome: Successfully transformed the raw dataset into an analysis-ready dataset, enabling effective exploration of patient flow, waiting time patterns, and operational performance.

2. SQL-Based Analysis

Objective

Analyse Emergency Department operations using SQL to identify patient flow patterns, waiting time trends, departmental performance, and admission behaviour.

- **Analysis 1: Total Patient Volume**

Methodology:

Calculated the total number of patient visits recorded in the Emergency Department.

Result:

A total of **9,216 patient visits** were analyzed.

Business Insight:

The Emergency Department operates in a high-volume environment, highlighting the need for efficient patient flow management and resource allocation.

- **Analysis 2: Average Waiting Time**

Methodology:

Calculated the average waiting time experienced by patients before receiving medical attention.

Result:

The average patient waiting time was **35.26 minutes**.

Business Insight:

While waiting times remain manageable, there is an opportunity to improve operational efficiency and reduce delays during busy periods.

➤ Analysis 3: Department Performance Analysis

Methodology:

Compared average waiting times across different referral departments to identify potential bottlenecks.

Result:

Neurology recorded the highest average waiting time (**36.80 minutes**), followed by **Physiotherapy (36.57 minutes)**.

Business Insight:

The difference between departments was relatively small, suggesting that delays are spread across the Emergency Department rather than being concentrated in a single area.

➤ Analysis 4: Peak Arrival Hour Analysis

Methodology:

Analyzed patient arrival patterns by hour to identify periods of increased demand.

Result:

The highest patient arrivals occurred at:

- 11:00 PM (436 patients)
- 07:00 AM (415 patients)
- 01:00 PM (410 patients)

Business Insight:

Patient arrivals were distributed throughout the day, with multiple demand peaks rather than one specific rush hour.

➤ Analysis 5: Severity Level vs Waiting Time

Methodology:

Evaluated whether patient severity levels influenced waiting times.

Result:

Average waiting times remained consistent across all severity categories, ranging between **35.03 and 35.45 minutes**.

Business Insight:

Severity level had minimal impact on waiting time, indicating a relatively uniform treatment process across patient groups.

➤ Analysis 6: Admission Rate Analysis

Methodology:

Calculated the percentage of patients admitted versus discharged.

Result:

- Admitted: **50.04%**
- Discharged: **49.96%**

Business Insight:

The nearly equal distribution demonstrates the Emergency Department's dual role in providing immediate treatment while supporting inpatient admissions.

➤ **Analysis 7: High-Wait Patient Analysis**

Methodology:

Identified patients who waited more than 45 minutes before receiving care.

Result:

A total of **2,750 patients** experienced waiting times exceeding 45 minutes, representing **29.84%** of all visits.

Business Insight:

Nearly 3 out of every 10 patients experienced extended waiting periods, indicating opportunities for process optimization and workload balancing.

➤ **Analysis 8: Busiest Day Analysis**

Methodology:

Identified the dates with the highest patient volumes.

Result:

The busiest day recorded **30 patient visits**, while the remaining top 10 days ranged between **25 and 28 visits**.

Business Insight:

Patient demand remained relatively stable throughout the analysis period, with no major spikes in Emergency Department utilization.

➤ **Analysis 9: Department Ranking Analysis**

Methodology:

Ranked departments based on average waiting times using SQL window functions.

Result:

Rank	Department	Average Wait Time
1	Neurology	36.80 min
2	Physiotherapy	36.57 min
3	Gastroenterology	35.83 min

Business Insight:

Neurology emerged as the department with the highest average waiting time and should be monitored for potential workflow improvements and resource allocation adjustments.

3. Dashboard Development & Visualization

Objective

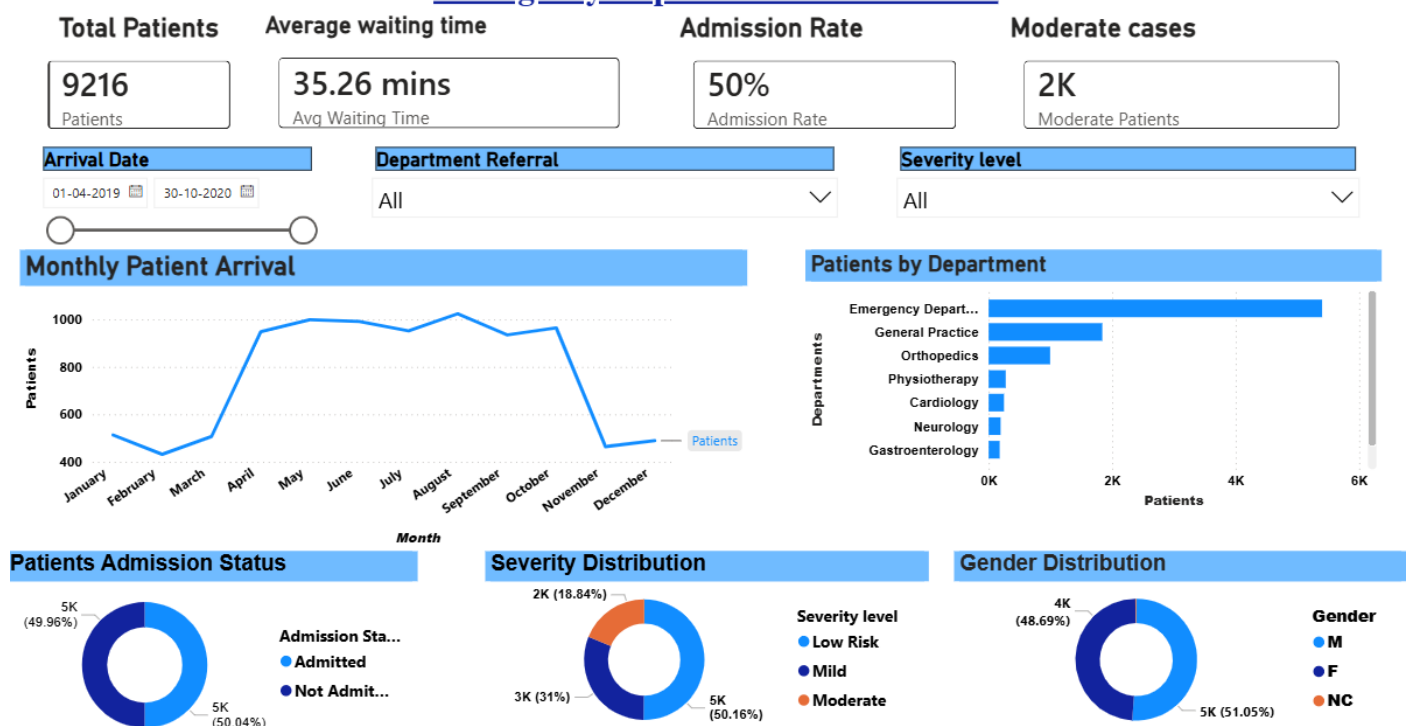
Develop an interactive Power BI dashboard to monitor Emergency Department performance, analyze patient flow, identify operational bottlenecks, and forecast future waiting time trends.

➤ Dashboard 1: Emergency Department Overview

Methodology

Created KPI cards and summary visualizations to provide a comprehensive view of Emergency Department operations, including patient volume, admission patterns, referral distribution, and patient demographics.

Emergency Department Performance



Result

- Analyzed **9,216** patient visits.
- Average patient waiting time was **35.26 minutes**.
- Admission rate remained balanced at **50%**.
- Most patients were classified under the **Low-Risk** severity category.
- Patient referrals were distributed across multiple specialty departments, with General Practice receiving the highest volume of referrals.

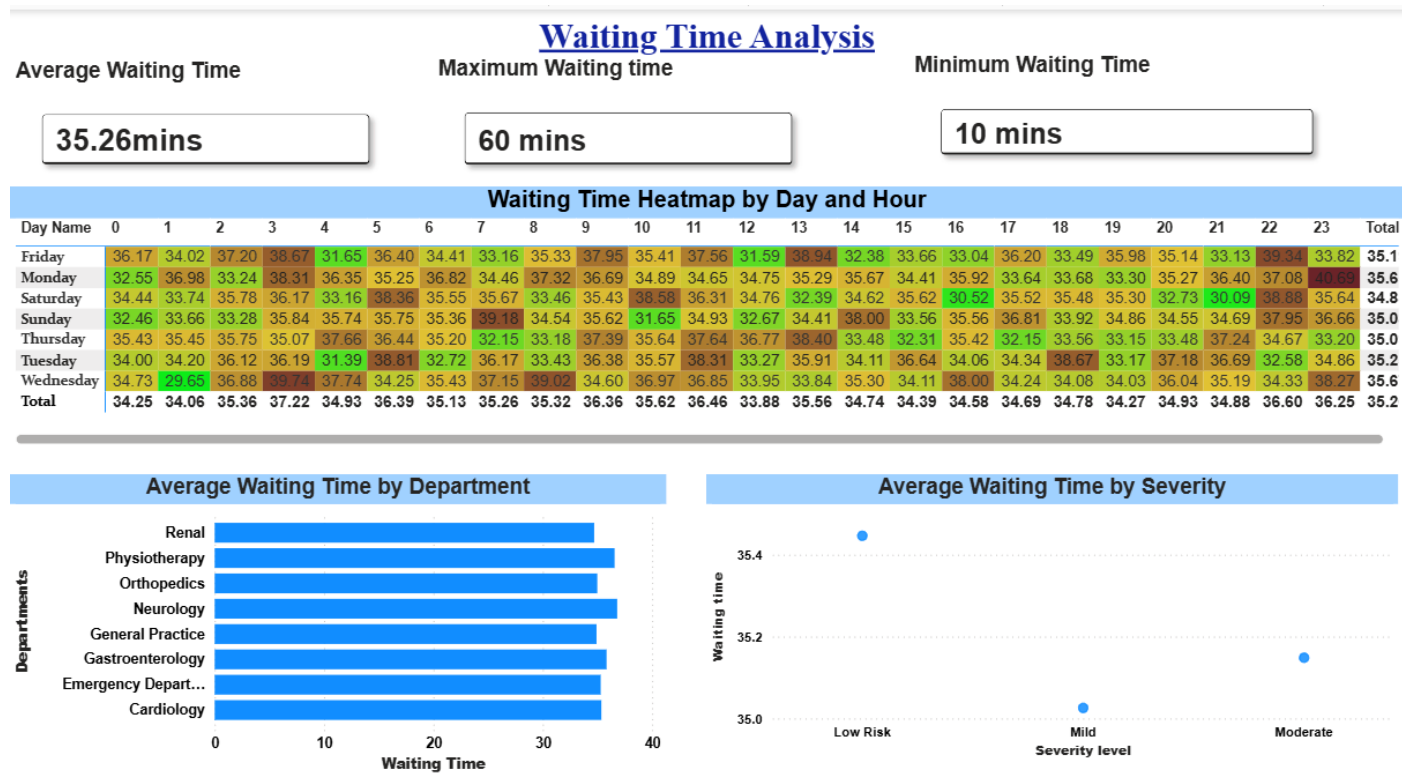
Business Insight

The Emergency Department managed a substantial patient load while maintaining stable admission patterns. Referral trends highlight the importance of efficient coordination between the Emergency Department and specialty departments.

➤ Dashboard 2: Waiting Time Analysis

Methodology

Analyzed waiting times by referral department, severity level, day of the week, and arrival hour to identify patterns contributing to patient delays.



Result

- Average waiting time was **35.26 minutes**.
- Maximum waiting time recorded was **60 minutes**.
- Among referral departments, **Neurology** reported the highest average waiting time, followed by Physiotherapy.
- Waiting times remained relatively consistent across severity categories.
- Certain day-hour combinations experienced slightly higher waiting periods than others.

Business Insight

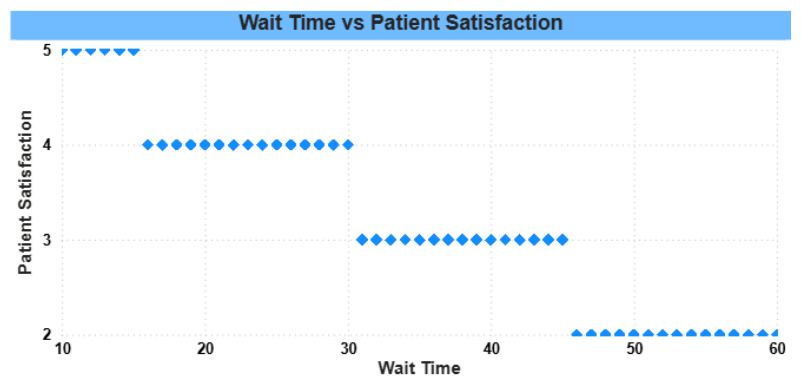
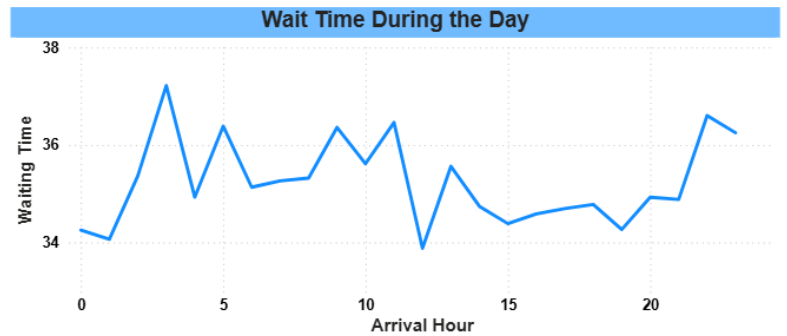
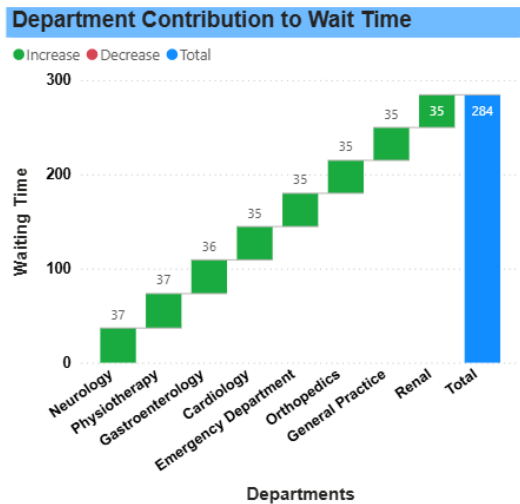
Patient delays appear to be influenced more by overall patient volume and referral workload than by severity level. Monitoring referral department performance can help reduce waiting times and improve service delivery.

➤ Dashboard 3: Root Cause Analysis

Methodology

Used decomposition analysis, departmental contribution metrics, hourly waiting trends, and patient satisfaction measures to identify factors driving Emergency Department delays.

Factors Driving Emergency Room Delays



Result

- Neurology contributed the highest average waiting time among referral departments.
- Waiting times fluctuated throughout the day, indicating periods of increased operational pressure.
- Higher waiting times were associated with lower patient satisfaction scores.

Business Insight

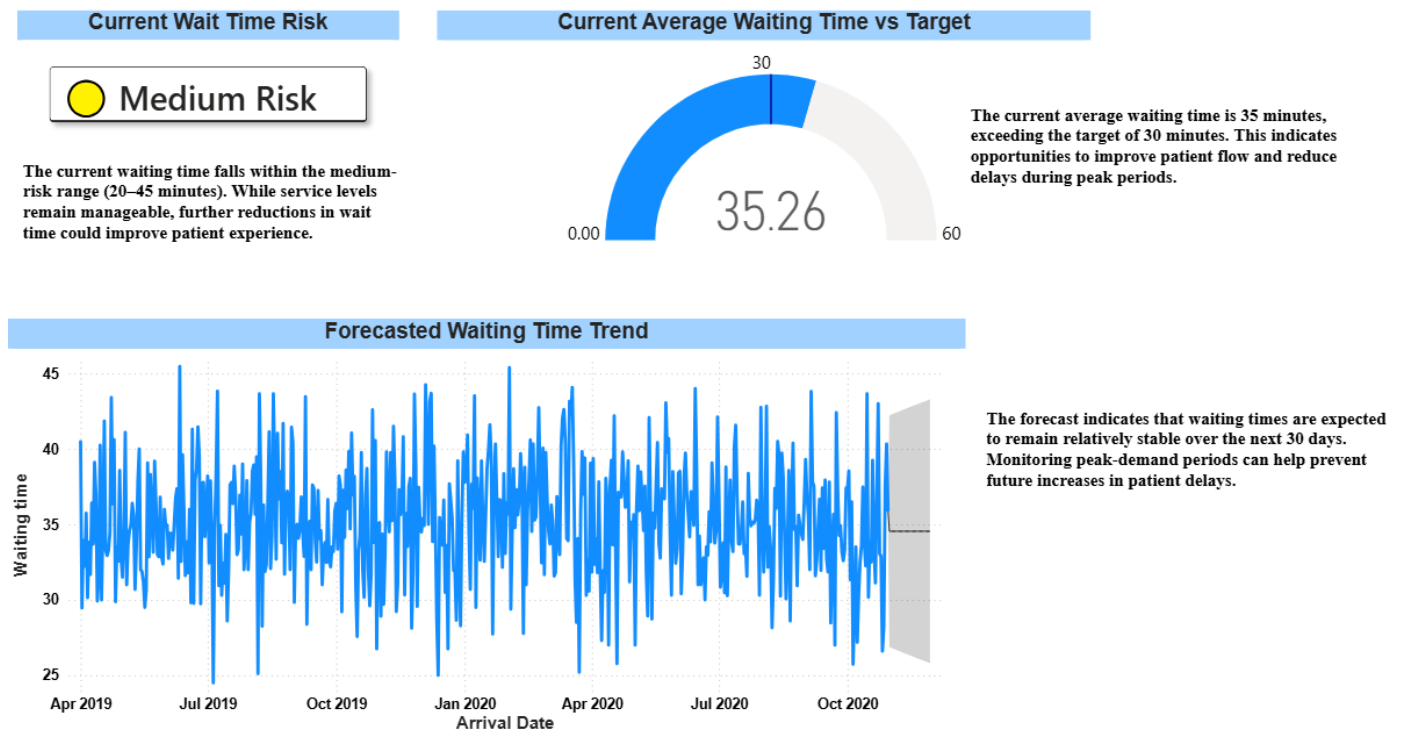
Referral department workload and peak arrival periods are key contributors to waiting time variations. Improving resource allocation during high-demand periods could enhance patient experience and operational efficiency.

➤ Dashboard 4: Waiting Time Forecasting & Performance Monitoring

Methodology

Applied Power BI forecasting techniques to historical waiting time data to estimate waiting time trends for the next 30 days and compare current performance against operational targets.

Predictive Wait Time Analysis & Performance Monitoring



Result

- Current average waiting time was **35.26 minutes**.
- The hospital's target waiting time was **30 minutes**.
- Current waiting time status was classified as **Medium Risk**.
- Forecast results indicated that waiting times are expected to remain relatively stable over the next 30 days.

Business Insight

Although waiting times remain manageable, they currently exceed the desired target. Forecasting provides visibility into future performance and can support proactive staffing and operational planning decisions.

Conclusion

This project successfully analyzed **10,000 Emergency Department patient records** to evaluate patient flow, waiting time performance, admission patterns, and referral department workloads. Through data cleaning, SQL-based analysis, and interactive Power BI dashboards, key operational trends and factors influencing patient delays were identified.

The analysis revealed that the Emergency Department handled a high volume of patients, with an average waiting time of **35.26 minutes** and an admission rate of approximately **50%**. While waiting times remained relatively consistent across severity levels, certain referral departments, particularly **Neurology**, experienced higher average waiting times. Additionally, patient arrivals varied throughout the day, indicating the importance of aligning staffing and resources with demand patterns.

The waiting time forecast further indicated that patient waiting times are expected to remain stable over the next 30 days, providing hospital management with valuable insights for proactive planning and resource allocation.

Overall, this project demonstrates how data analytics can be leveraged to transform healthcare data into actionable insights. By identifying operational bottlenecks, monitoring performance metrics, and forecasting future trends, hospitals can make informed decisions to improve patient experience, optimize resource utilization, and enhance Emergency Department efficiency.

References

1. Kaggle Dataset Repository - *Hospital Emergency Room Dataset*
2. Microsoft Excel
3. Microsoft Power BI Documentation
4. MySQL Documentation